

Complex Number Operations

Adding, subtracting, and multiplying complex numbers

Directions

- Write every final answer in standard form $a + bi$, with a and b real numbers.
- Whenever a term i^2 appears, replace it with -1 before you combine like terms.
- If an answer is purely real or purely imaginary, say so by writing it with the missing part as 0 or by writing the single term (for example 29 or $6i$).
- Show the multiplying-out step, not just the final line.

1. Simplify: $(8 + 3i) + (2 + 9i)$

Answer: _____

2. Simplify: $(7 - 4i) - (2 + 6i)$

Answer: _____

3. Simplify: $(-3 + 5i) + (9 - 11i) - (4 + 2i)$

Answer: _____

4. Multiply and write the result in standard form: $3i(4 - 5i)$

Answer: _____

5. Multiply and write the result in standard form: $(2 + 3i)(4 - i)$

Answer: _____

6. Simplify: $(10 - 2i) - (-3 + 7i)$

Answer: _____

7. Multiply: $(5 - 2i)(5 + 2i)$

Answer: _____

The two factors are *conjugates*. Describe what kind of number the product turned out to be, and why the i terms cancelled.

Answer: _____

8. Expand and simplify: $(3 + 4i)^2$

Answer: _____

9. Multiply and write the result in standard form: $(-6 + i)(2 - 5i)$

Answer: _____

10. Simplify: $2(4 - 3i) - 3(1 + 5i)$

Answer: _____

11. Simplify: $(2 - 3i)^2 + (2 + 3i)^2$

Answer: _____

12. Let $z = 3 - 2i$ and $w = -1 + 5i$.

- (a) Compute $zw - 2z$ and write the result in standard form.

Answer: _____

- (b) In one or two sentences, explain why the product of two numbers that each carry an imaginary part can end up with a real part that neither factor had. Point to the step where i^2 is replaced by -1 .

Answer: _____